

Password Authenticated Key Agreement (PAKE), a form of Diffie-Hellman is used to prevent man-in-the-middle attacks which is like eavesdropping. Other variants of Diffie-Hellman, such as STS (Station-to-Station Protocol) may be used instead to avoid these types of attacks in an efficient manner. So in a way the Diffie-Hellman algorithm is the basis for currently used cryptographic techniques since this algorithm worked very well when generating cryptographic keys to encrypt data communicated over a public channel.

DNA Fingerprinting

Invented accidentally by Alec Jeffreys when he was studying gene evolution, much to the delight of the CSI television show. Its first use was in confirming the identity of a British boy whose family was originally from Ghana. Jeffreys realised the enormity of what he just invented and so he left whatever he was doing and went head on into research towards other applications of DNA fingerprinting.

Since about 99% of the DNA is identical in humans, the variable part is assessed; this uniqueness is the base of DNA fingerprinting.

The first use of it in a criminal investigation was in the case of the murder of two girls. A Richard Buckley was arrested for the murder of both the girls but had confessed to only one. Using DNA fingerprinting the true murderer of the second girl was caught.

